

# **BANK MARKETING CAMPAIGN CASE STUDY**

## **1. Introduction**

Term deposits are one of the most common investment products offered by banks. They allow customers to deposit funds for a fixed period in exchange for a guaranteed interest return. For financial institutions, increasing term deposit subscriptions is important because it helps secure stable funding and improve long-term profitability.

Banks often use direct marketing campaigns, such as phone calls and other communication channels, to promote term deposit products. However, not all customers respond positively to these campaigns. Understanding the factors that influence customer subscription behavior can help banks improve campaign effectiveness, reduce marketing costs, and target the most promising customer segments.

This case study analyzes a bank marketing dataset containing customer demographic information, financial characteristics, and campaign interaction history. The objective is to identify the key factors that influence term deposit subscriptions and provide actionable recommendations that can support data-driven marketing decisions.

## **2. Business Problem**

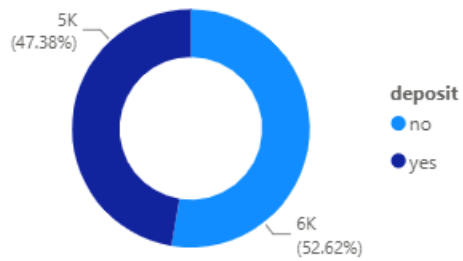
- Why do some customers subscribe to term deposits and others don't?
- Which customer segments should the bank target?

## **3. Dataset Overview**

- 11,162 rows
- 17 columns
- 0 missing values
- Target variable: deposit
- Numerical Variables: age, balance, duration, campaign, pdays, previous
- Categorical Variables: job, marital, education, housing, loan, contact, poutcome, deposit
- Summary statistics:

| <b>Metric</b>             | <b>Value</b> |
|---------------------------|--------------|
| Average Age               | 41.23 years  |
| Average Balance           | 1,528.54     |
| Average Campaign Contacts | 2.51         |
| Average Duration          | 371.99       |

### Subscription Distribution



## 4. Analysis Process

The analysis was conducted using SQL for data exploration and customer segmentation, while Power BI was used for visualization.

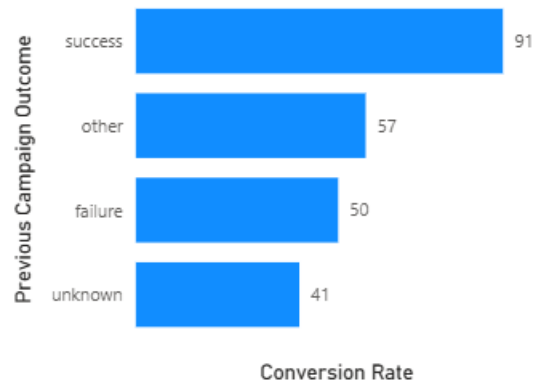
The following steps were performed:

- Data exploration and validation
- Missing value assessment
- Descriptive statistical analysis
- Customer segmentation
- Conversion rate calculations
- Identification of key subscription drivers
- Visualization of findings and business insights

## 5. Key Findings

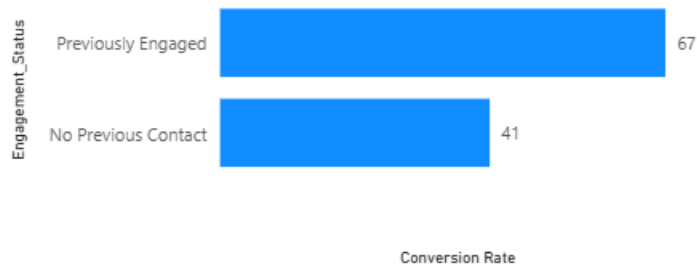
1. Previous Campaign Success Drives Future Subscription. Customers with a previous successful campaign outcome recorded a conversion rate of 91.32%, the highest among all customer segments.

Previous Outcome Conversion Rate(%)



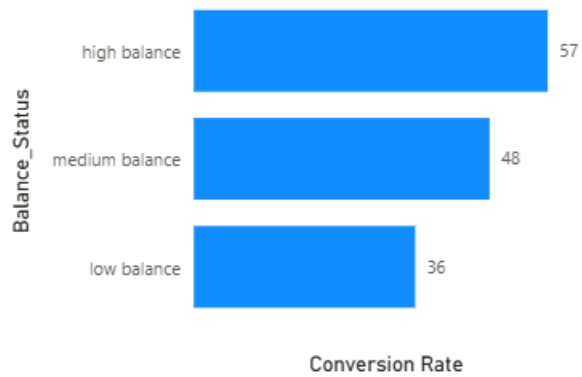
2. Previous engagement customers convert more (67.12% vs 40.65%)

Engagement Conversion Rate(%)

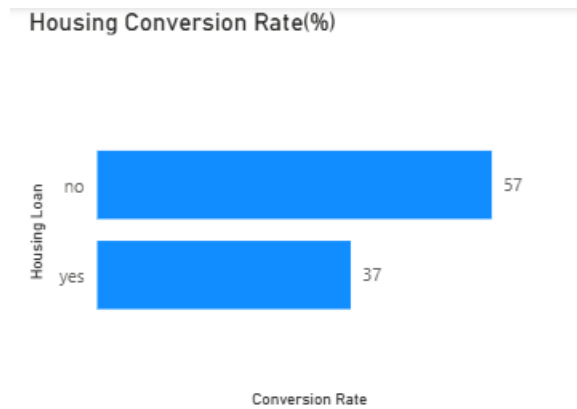


3. High-balance customers convert more (57.49%)

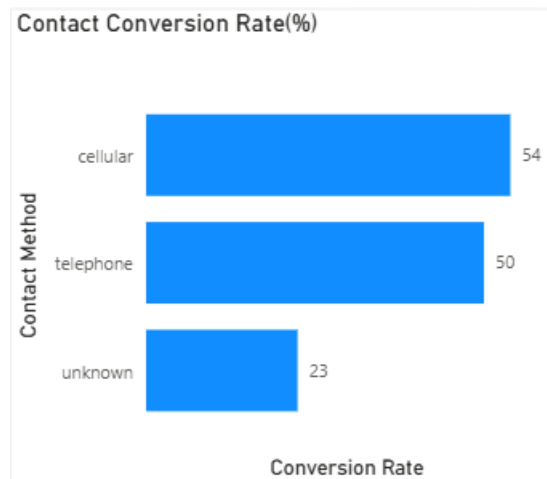
Balance Conversion Rate(%)



4. Customers without housing loans convert more (57.03%)



5. Cellular is the best communication channel (54.33%)



## 6. Business Insights

Previous campaign outcome is the strongest predictor of subscription behavior in the dataset. Customers who previously subscribed following a marketing campaign recorded a conversion rate of 91.32%, significantly outperforming all other customer groups. This suggests that historical campaign performance can be a powerful tool for identifying high-potential prospects.

Customer familiarity significantly improves subscription likelihood; however, excessive repeated contact appears to produce diminishing returns and may reduce campaign effectiveness.

Customer balance is a strong predictor of term deposit subscription behavior. Customers in the high-balance segment recorded the highest conversion rate (57.49%), while customers in the low-balance segment recorded the lowest conversion rate (35.96%).

Housing loan status appears to strongly influence term deposit subscription behavior. Customers without a housing loan recorded a conversion rate of 57.03%, compared to only 36.64% among customers with a housing loan. This suggests that existing mortgage obligations may reduce customers' willingness or ability to invest in term deposit products.

Communication channel influences campaign effectiveness. Customers contacted through cellular channels recorded the highest subscription rate (54.33%), slightly outperforming telephone contacts (50.39%). Records with unknown contact methods achieved substantially lower conversion rates (22.59%), highlighting the importance of maintaining complete contact information.

## **7. Recommendations**

1. Prioritize cellular communication for future marketing campaigns, as it achieved the highest customer conversion rate. Improve customer contact information management to reduce records with unknown contact methods, which are associated with significantly lower campaign success rates. Continue using telephone campaigns as a secondary channel, as their performance remains relatively strong and close to cellular campaigns.

2. Prioritize customers without housing loans in future term deposit campaigns, as they demonstrate substantially higher subscription rates. Bundle mortgage and savings products where appropriate, helping customers build savings while managing loan commitments.

3. Offer personalized investment recommendations and premium savings products to high-balance customers, as this segment shows the strongest propensity to subscribe.

Develop entry-level savings and investment products targeted at low-balance customers, who represent a lower-converting segment but may offer long-term growth opportunities.

4. Focus on ensuring customers receive at least one meaningful engagement rather than repeatedly contacting the same individuals.

Develop customer nurturing programs that establish familiarity with the bank and its products.

Allocate campaign resources toward expanding engagement across more prospects instead of concentrating excessive efforts on a small number of customers.

5. Develop loyalty and retention campaigns specifically targeting customers with a history of successful subscriptions.

Continue engaging customers with previous campaign failures, as they still demonstrate higher conversion rates than completely unknown prospects.

## **8. Conclusion**

This analysis sought to identify the factors influencing customer subscription to term deposit products. The findings revealed that previous campaign outcomes, prior customer engagement, account balance, housing loan status, and communication channels were among the strongest predictors of subscription behavior.

Customers with successful previous campaign outcomes demonstrated exceptionally high conversion rates, while customers with stronger financial positions were generally more likely to subscribe. The analysis also showed that maintaining customer relationships and using effective communication channels can significantly improve campaign performance.

By leveraging customer data and historical campaign performance, the bank can improve marketing efficiency, optimize resource allocation, and increase term deposit subscription rates